Refining the ETH contraction

A residue-aware Cauchy basis, sharper packing,
and a cheaper recursive precision schedule

Research prepared for Vladimir Reshetnikov

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Abstract

We review all seven ETH proposal archives present in the pinned ProveIt snapshot, together with the cited OpenAI preprint and the separately supplied 49-page manuscript. We obtain a new explicit contraction certificate

$$(s+\mathcal{L})^{O(1)}\,2^{(133.20737249245\ldots+o(1))s/\sqrt{\log_2 s}}$$

for the same normalized Boolean tensor interface used by the strongest previous explicit schedule. Its displayed coefficient was $648.4410666487\ldots$; combining that schedule with the prior factorial-Cauchy basis already gives $374.3776243832\ldots$. Our proof improves both. The contributions are an asymptotically sharp use of the packing load bound, an additive precision guard replacing doubled packet exponents, and an exact Cauchy separation that exploits a missing residue class in the actual rows. The last changes the mode-weight slope from 3 to 64/21 without adding mode families. The new local lemmas are proved here; the complete contraction conclusion uses the unchanged deformation and recursive identities of the preprint. This is an improvement of a specified exponential upper-bound coefficient. It does not establish a better logarithmic order for unrestricted 3-SAT, verify the entire preprint, or compare unknown constants in the separate 49-page algorithm. Exact executable checks accompany the proofs.

1 The result and the comparison standard

Let n be the number of declared Boolean variables, m the input clause count, and k a fixed clause width. The requested case is $k = 3$. All algorithms discussed here count satisfying assignments exactly and therefore also decide satisfiability, subject to their stated hypotheses. Polynomial factors include reading the encoded input. Unless explicitly written as ln, logarithms are base two.

The strongest statement in the cited OpenAI manuscript is its quantitative Appendix D, incorporated into its main theorem:

$$T_{\text{general}}(n,m) = (n+m)^{O_k(1)}2^{O_k(n\log\log n/\sqrt{\log n})}, \tag{1}$$

$$T_{\text{fixed}}(n,m) = (n+m)^{O_{k,D}(1)}2^{O_{k,D}(n/\sqrt{\log n})} \qquad (m \le Dn). \tag{2}$$

Both use polynomial space [1, 3]. The earlier $(\log\log n)^2$ numerator in the body and the reviewed Lean statement is a weaker endpoint. Reproducing Appendix D would not improve the strongest existing proposal.

Our principal theorem concerns the contraction subroutine with a normalized wire budget s and description length $\mathcal{L}$; its full interface is in Section 3. Write $\lambda = \lceil \log_2 s \rceil$. The result is

$$T_{\text{new}}(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)}2^{(C_*+o(1))s/\sqrt{\lambda}}, \qquad C_* = \frac{9\pi}{4\ln 2}\sqrt{\frac{1365}{8}} = 133.20737249245\ldots. \tag{3}$$

The $o(1)$ is uniform over instances satisfying fixed interface constants. For example, $C_* + o(1)$ can be replaced by 134 for all sufficiently large s. The threshold is not asserted to be practical.

Specified contraction certificate	Coefficient of $s/\sqrt{\log_2 s}$
Previous retuned schedule and improved taper [5]	648.4410666487
That schedule composed with the existing factorial-Cauchy basis	374.3776243832
Boundary infimum of the earlier optimized family, with that basis	368.6540181167
New packing and additive guard, existing slope-3 basis	134.2604115868
New packing, additive guard, and the residue-aware basis	133.2073724925

Table 1: Matched normalized interfaces and bit-cost accounting. These are upper-bound coefficients, not measured times or lower bounds for earlier algorithms. Rows two and three compose the earlier schedules with the prior basis; they are not quoted standalone results.

The new certificate is smaller than every explicit comparable contraction certificate established in the seven reviewed packets, including the displayed earlier schedule composed with factorial-Cauchy modes. For completeness, the earlier schedule packet also optimizes its continuous parameter family: after composition with that basis its boundary infimum is approximately 368.654018, near $r = 251$. Strict packing slack approaches rather than attains that boundary. The new coefficient is smaller than this comparison too. Its logarithmic order is still $s/\sqrt{\log s}$. Consequently (1)–(2) remain the established SAT orders. Several packets propose faster orders under additional unproved compression or summation assumptions. We do not claim to surpass those hypothetical algorithms, to prove their missing assumptions, or to establish a runtime inequality between implementations whose constants have not been specified.

2 Repository review and provenance

The ProveIt snapshot is

`aa6933442eb936962de79ba6d321cd0391c450bd`.

The cited OpenAI source snapshot is

`7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`.

The three relevant intake commits below are dated 26 October 2026 in Pacific time. Their contents, including the ZIP members rather than only commit messages, were inspected. The archive manifest records immutable URLs, Git object identifiers where available, byte sizes, and SHA-256 fingerprints. No repository files or remote branches were changed.

A subsequent ETH placement commit, `2b925d8800`, moved these materials into `Computability/SubexponentialSAT/Research`. That commit and the new project README were also reviewed. All 119 placed packet files were independently matched against the downloaded archive members by their Git blob hashes; the mathematical corpus is unchanged. The report retains the earlier immutable archive links, and the provenance manifest records the current paths as well [12].

Intake commit	ETH-related additions
`a505bd768b`	Factorial-Cauchy modes; contraction/taper schedule; density-aware counting; original `paper.pdf`; separate `eth.pdf`.
`64b3202ee8`	Precision refinement; scalar parity/refinement packet; second density refinement.
`8afdbbea48`	Asymptotic-analysis packet with conditional degree-preserving compression and restricted method barriers.

2.1 What each prior packet actually establishes

Packet	Established contribution within its stated model	Stronger unresolved target
Factorial Cauchy [4]	Integral exact separation with weights $1 + b + 2v_2(b!) = 3b + O(\log b)$; matched entropy coefficient reduced by $1/\sqrt{3}$.	Better log-power needs additional structure.
Contraction improvements [5]	Exact Hermite condition count, leading precision $6MD/u$, schedule coefficient $648.44\ldots$, and streamed packet coefficient extraction.	Same global log-power.
SAT density improvement [6]	For fixed-polylogarithmic density B, exponent $O(n\log(2+B)/\sqrt{\log n})$, with explicit row-construction costs.	Uniform compression at arbitrary density.
ETH density refinement [7]	A related density-sensitive construction and checks of retuning and interpolation losses.	No proved unrestricted lower log-order.
Precision refinement [8]	Exact node-dependent precision and optimal taper analysis for its loss functional.	A structured-mode summation hypothesis would give an $s/\log s$ scale.
Scalar parity refinement [9]	Removes the nilpotent variable in that reduction and lowers incidence degree from $k(a+2n)$ to $k(a+n)$.	A second contraction needs a compact high-degree-group factorization.
Asymptotic analysis [10]	Restricted counting and encoding barriers; precise interface hypotheses.	A degree-preserving compiler would yield $O(n\sqrt{\log\log n/\log n})$; a stronger compiler would yield $O(n/\sqrt{\log n})$.

These results are not seven independent proofs of a new log-power. In particular, an exact finite representation of a mode sum does not supply an efficient contraction of that representation. Likewise a retained incidence-degree cap does not automatically bound the multilinear degree of a binary subset index. The reviewed packets generally state these boundaries explicitly; this report preserves them.

2.2 The separate 49-page manuscript

The file `eth.pdf`, titled *A $2^{o(n)}$ time algorithm for 3-SAT*, attributes itself to Claude/Anthropic and is dated 26 October 2026, 3:13 PM PT. It is a different construction, based on decompositions of a crossing, recursive clique algorithms, and partition dimensions. Its Theorem 1 has exponent

$$O\left(\min\left\{k^2 n\frac{\log\log n}{\sqrt{\log n}},\ km\frac{1}{\sqrt{\log m}}\right\}\right)$$

in its stated computational model. For fixed k this has the same leading orders as the OpenAI manuscript. Its Appendix B supplies $\rho = 1/\sqrt{2}$ to the recurrence in Appendix A, which yields the logarithmic exponent $\log_2(1/\rho) = 1/2$ [11]. It does not provide a coefficient directly comparable with

Table 1's normalized Boolean-wire interface. We therefore record its rate and distinct model without claiming that (3) is faster than its complete algorithm. Neither that manuscript's Lean development nor the OpenAI Lean development was rebuilt for this report.

3 The contraction interface and inherited machinery

The normalized input is a bipartite Boolean tensor multigraph with at most s edges, polynomially many vertices in s, and maximum degree at most $\Delta = \left\lceil s^{9/10} \right\rceil$. Parallel edges have separate ports. The entries belong to

$$\mathcal{B}_{P_0,\ell} = (\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell), \qquad 1 \le P_0, \ell \le s.$$

Given its immutable description and a Boolean assignment to its ports, an original tensor entry is computable in $(s+\mathcal{L})^b 2^{As^{9/10}}$ bit operations and $(s+\mathcal{L})^b$ bits of space for fixed A, b. Constants in a more general Cs interface can be absorbed into s; the explicit coefficient then refers to that rescaled s, not directly to n.

The source extends scalars to an unramified extension $\mathcal{O}$ of $\mathbb{Z}_{(2)}$, with sufficiently many distinct residue-field elements for unit-difference interpolation. Valuations are normalized by $v_2(2) = 1$ and are taken coordinatewise in the truncated V ring. All precisions and extension dimensions used below are polynomial in s.

At a recursive call with M wires, a packing creates packets of $L = 2^a$ slots. A Cauchy polynomial matrix $W(T)$ has $W(0) = I$ and degree $2L-1$ per entry. Applying the multilinear row tensors to $W(T)X$ gives a deformation $F(T)$ of degree at most $M(2L-1)$ whose value at zero is the desired contraction. At

$$T = t + \alpha\zeta, \quad \alpha = 2^{a+1}, \quad t \in \{-L, \dots, L-1\}, \quad \zeta \in \mathcal{O},$$

the local kernel is integral. For retained $i = v_2(t)$, grouping vertices by label coordinate i makes $W(t)$ local. The remaining crossing interactions have an integral separated expansion into modes.

The source's recursive identity sums over mode exponent tuples and takes a product of one smaller child contraction per group. Its validity needs only integral target coefficients with known valuations and source features confined to a single group. These are exactly the properties preserved by our new basis. We retain the deformation, recovery identity, and child construction [2].

Immutable precision semantics. A tensor is tagged with its creation precision. Whenever a later call uses it at a larger precision, its entry is first evaluated at its creation precision and the same integer coordinate representatives are lifted. Canonical lifting is not asserted to preserve multiplication or addition. New tensors retain their creation-time descriptions. This rule is necessary for the source's induction and is unchanged here.

4 The taper improvement already available

We use the strongest established precision refinement rather than counting it as a new contribution. Put $q_i = h(u-i)$ for $0 \le i < u$, and $q_i = 0$ otherwise, where $1 \le u \le a$. There are $L/2^i$ nonzero grid nodes of valuation $i < u$. Consequently the exact number of Hermite conditions is

$$\sum_t q_{v_2(t)} = hL(2u - 2 + 2^{1-u}). \tag{4}$$

Use the integer choice

$$h = \left\lceil \frac{M(2L-1)+1}{L(2u-2+2^{1-u})} \right\rceil. \tag{5}$$

It supplies more conditions than the deformation degree. Whenever $u \longrightarrow \infty$, $u/M \longrightarrow 0$ and $L \longrightarrow \infty$ uniformly across active calls,

$$h = (1+o(1))M/u. \tag{6}$$

For $D_i = L/2^i$, the exact loss used in the precision packet is

$$B_i = D_i \sum_{j=1}^{a-i} \frac{j}{2^j}(q_i - q_{i+j}).$$

The sufficient working precision is

$$P_t = p + B_i + (q_i - 1)(a+1-i), \tag{7}$$

where p is the call's output precision. The Hermite multiplier has valuation $-B_i$, and evaluation of a coefficient of order $j < q_i$ at $-t/\alpha$ loses $j(a+1-i)$ powers of two. These statements prove (7) by bounding the contribution of each input error separately. The truncated reciprocal with constant term one is integral, so forming it loses no precision [8].

Since $q_i - q_{i+j} \le hj$ and $\sum_{j\ge1} j^2/2^j = 6$, we have $B_i \le 6hD_i$. Once the inherited precision and the final jet-evaluation term are negligible, this gives

$$P_t \le (6+o(1))MD_i/u. \tag{8}$$

The next sections establish that negligibility for the new schedule. The exact optimal-taper result in the earlier precision packet explains why changing only this taper does not supply another log-order gain.

5 New contribution I: keep the sharp packing bound

Let r be a fixed integer, let each vertex label belong to $[r]^a$, and put $N = r^{2a}$, the number of ordered pairs of labels. The source's conditional-expectation argument produces label-class loads satisfying

$$\max_{\gamma} X_\gamma \le M/N + \sqrt{2M\Delta}.$$

Padding every label-pair class to the maximum load rounded up to a multiple of L therefore gives

$$H_{\mathrm{pad}} \le M/N + \sqrt{2M\Delta} + L. \tag{9}$$

Define

$$\epsilon_s(M,a) = \sqrt{\frac{2N^2\Delta}{M}} + \frac{LN}{M}.$$

Multiplication of (9) by N yields $H_{\mathrm{pad}}N \le (1+\epsilon_s)M$. Thus the same deterministic packing has the following sharper consequences:

$$\#\text{packets} \le (1+\epsilon_s)M/L, \tag{10}$$

$$\#\text{ports per side in a group} \le (1+\epsilon_s)M/r, \tag{11}$$

$$M_{\mathrm{child}} \le 2(1+\epsilon_s)M/r, \tag{12}$$

$$c_t := \#\text{column-tile families} \le 2(1+\epsilon_s)M/D_i. \tag{13}$$

The last line uses 2^{i+1} possible tiles per packet. It counts mode families, each of which has one variable per mode index. In contrast, the source and earlier schedules replace the right side of (9) by $3M/N$, giving $6M/D_i$ families and $6M/r$ child wires. Under strict-power packing slack, $\epsilon_s = o(1)$ uniformly, so that factor three can be removed from the leading constant. This is a refinement of the proven load estimate; the packing algorithm itself does not change.

6 New contribution II: an additive precision guard

Take

$$\begin{aligned} \lambda &= \lceil \log_2 s \rceil, \quad r = 54, \quad H = \lceil s/\lambda \rceil, \quad g = \left(\lceil \log_2 \lambda \rceil\right)^2, \\ u_d &= \left\lfloor \frac{\lambda}{260 \cdot 9^d} \right\rfloor, \qquad a_d = \sum_{j=0}^{d} u_j + (d+1)g, \qquad a_{-1} = 0. \end{aligned} \tag{14}$$

An active call has $M > H$; smaller calls are evaluated directly. As in the source, bounded exceptional input sizes are handled by exhaustive summation. The relevant integer inequalities can also be checked before recursion. The following proof shows that the recursive regime applies uniformly once s exceeds a fixed threshold.

6.1 Packing slack and a noncircular induction

Write $A_0 = 9/2080$. At depths $O(\log \lambda)$,

$$a_d \le A_0 \lambda + O((\log \lambda)^3). \tag{15}$$

The exact integer inequality $54^{25} < 2^{144}$ gives

$$\frac{9}{10} + 4A_0 \log_2 54 < \frac{9}{10} + 4\frac{9}{2080}\frac{144}{25} = \frac{3249}{3250} < 1. \tag{16}$$

Therefore $N^2\Delta = s^{1-\Omega(1)}$ and $LN = s^{1-\Omega(1)}$, uniformly at those depths. Since $M > s/\lambda$, we get $\epsilon_s = s^{-\Omega(1)}$. Also $2L \le \Delta$ eventually.

To justify the depth range without assuming the desired shrinkage, start with the provisional invariant $M_d \le s/16^d$. It holds at the root. At an active call it bounds d by $\log_{16} \lambda + O(1)$, so (15) and the packing estimates already apply. Because $2/54 < 1/16$, (12) implies the invariant at the next depth for all sufficiently large s. This closes the induction. Since $9 < 16$, this provisional argument also proves that the retained u_d tend to infinity uniformly.

We may now use the sharp ratio. Multiplying $M_{d+1} \le (1+\epsilon_s)M_d/27$ over $O(\log \lambda)$ depths, with a uniform negative-power bound on ϵ_s, gives

$$M_d \le (1 + o(1))s/27^d. \tag{17}$$

At active calls this implies $d \le \log_{27} \lambda + O(1)$ and

$$u_d = \Omega(\lambda^{1/3}), \qquad u_d/M_d = o(1). \tag{18}$$

Floors in (14) hence cause only uniform relative $o(1)$ errors. Both (6) and the sharp-packing estimates hold throughout the recursion.

6.2 The guard absorbs inherited precision

The critical identity is exact:

$$a_d - u_d = a_{d-1} + g.$$

For every retained valuation $i < u_d$,

$$D_i = 2^{a_d - i} \geq 2^{a_{d-1}+g+1}. \tag{19}$$

The source precision argument, with the smaller increments from (7), gives

$$p \leq s\lambda^{O(1)} 2^{a_{d-1}}.$$

This follows directly by summing the increments along the path; each earlier increment is at most $s\lambda^{O(1)}2^{a_{d-1}}$, and the path has $O(\log \lambda)$ levels. Hence, at an active call,

$$\frac{p}{MD_i/u_d} \leq \lambda^{O(1)} 2^{-g} = o(1).$$

Also

$$\frac{hu_d(a_d+1)}{MD_i/u_d} = O\left(\frac{u_d(a_d+1)}{D_i}\right) \leq \lambda^{O(1)}2^{-g} = o(1).$$

Thus (8) follows uniformly. The additive guard contributes only $O((\log\lambda)^3) = o(\lambda)$ to the packet exponent. It replaces the earlier choice $a_d = 2\sum_{j\leq d} u_j$, whose extra full copy of the retained-level budget is unnecessary.

6.3 Sizes and entry costs

All packet and label spaces remain polynomial in s by (15). The deformation degree is $O(sL) = s^{1+A_0+o(1)} < s^4$, so the source's residue field of size $2^{4\lambda}$ still supplies enough interpolation points. All precisions are polynomial in s. New packet degrees are at most $2L \leq \Delta$, and old vertex degrees are unchanged. Even the source's original packet-entry procedure costs only $\exp(O(L\log s)) = \exp(o(s^{9/10}))$ here. The previously proved streamed finite-difference routine can be retained, but is not needed to justify our leading coefficient.

7 A residue-aware Cauchy separation

This section changes the separated basis used for crossing entries in one contraction call. It does not assume an unproved faster summation routine. The new basis has the same source and target groups as the previous factorial-Cauchy basis, but its guaranteed valuation grows with slope 64/21 instead of 3.

7.1 The actual row domain

Let $\mathcal{O}$ be the unramified coefficient ring used in the source construction, with valuation $\nu_2(2) = 1$. In particular, its reduction modulo two can be a proper extension of $\mathbb{F}_2$. Set

$$L = 2^a, \quad i = \nu_2(t) < a, \quad \alpha = 2^{a+1}, \quad T = t + \alpha\zeta, \quad \zeta \in \mathcal{O}, \qquad J = 2^{a-i-1}.$$

A column tile has positions $y = y_0 + 2^{i+1}j$, where $0 \leq y_0 < 2^{i+1}$ and $0 \leq j < J$. For a crossing row x, the bits in position i of x and y_0 are equal. Define

$$d_0 = t - y_0 + x, \qquad d = d_0 + \alpha\zeta, \qquad z_x = \frac{2^{i+1}}{d}, \qquad w_x = \frac{z_x}{2}, \qquad R = a + 1 - i.$$

The source's crossing condition gives $d_0 \neq 0$ and $\nu_2(d_0) \leq i$. With

$$\mathscr{C}(T) = \prod_{q=-L}^{L-1} (T-q), \qquad c_x = \frac{\mathscr{C}(T)}{\mathscr{C}'(0)d},$$

the kernel is

$$W_{xy}(T) = \frac{c_x}{1 - z_x j}, \qquad z_x \in 2\mathcal{O}, \quad \nu_2(c_x) \geq 1. \tag{20}$$

The last lower bound may be sharpened to R, but an offset of one already suffices for the comparison below.

Lemma 7.1 (A missing residue class)**.** *Put $w_0 = 2^i/d_0 \in \mathbb{Z}_{(2)}$. Then*

$$w_x - w_0 \in 2^R\mathcal{O}.$$

If $i \geq 1$, define $c \in \{2, 6\}$ by

$$c = \begin{cases} 6, & \text{bit } i-1 \text{ of } y_0 \text{ is zero,} \\ 2, & \text{bit } i-1 \text{ of } y_0 \text{ is one.} \end{cases}$$

For $R \geq 3$, the residue of w_x modulo eight lies in the prime subring and is different from $8 - c$. If $i = 0$, w_x is odd and either choice of c is valid.

Proof. Write $v = \nu_2(d_0) \leq i$. Adding $2^{a+1}\zeta$ preserves that valuation. Therefore

$$w_x - w_0 = -\frac{2^{i+a+1}\zeta}{d_0(d_0 + 2^{a+1}\zeta)}, \qquad \nu_2(w_x - w_0) \geq i + a + 1 - 2v \geq R.$$

This proves the prime-subring assertion when $R \geq 3$.

Only the case $\nu_2(w_0) = 1$ could give residue two or six. In this case $v = i - 1$. Write the lower i bits of x and y_0 as $x_<$ and $y_<$. Since their bits in position i agree, the other part of $x - y_0$ is divisible by 2^{i+1}. The difference $x_< - y_<$ lies strictly between -2^i and 2^i and, by $v = i - 1$, is an odd multiple of 2^{i-1}. It must be $+2^{i-1}$ when bit $i - 1$ of y_0 is zero, and -2^{i-1} otherwise. Since $t/2^{i-1} \equiv 2 \pmod 4$, the odd number $d_0/2^{i-1}$ is respectively three or one modulo four. Its inverse has the same residue, so w_0 is respectively six or two modulo eight. In all other valuation cases neither residue can occur. For $i = 0$, d_0 is odd, which proves the last statement. □

The exceptional value $R = 2$ gives $J = 1$. It has only the mode $b = 0$ and needs no residue exclusion.

7.2 An explicit sequence of auxiliary row points

Repeat the word of length 21

$$A\ B\ A\ B\ C\ A\ B\ A\ B\ C\ A\ B\ A\ A\ C\ A\ B\ A\ A\ C\ A.$$

At the kth occurrence of a letter, starting at $k = 0$, set

$$v_h = \begin{cases} 1 + 2k, & \text{letter } A, \\ 4k, & \text{letter } B, \\ c + 8k, & \text{letter } C, \end{cases} \qquad \rho_h = 2v_h. \tag{21}$$

The index h counts positions in the repeated word. Let A_b, B_b, C_b be the numbers of the three letters among its first b positions. Thus a complete block has counts $(11, 6, 4)$. Define

$$\eta_b = \min\left\{A_b + \nu_2(A_b!), \quad 2B_b + \nu_2(B_b!) + C_b, \quad 3C_b + \nu_2(C_b!) + B_b\right\}. \tag{22}$$

Lemma 7.2 (Factorials near the prime subring)**.** *If $q \in \mathcal{O}$, $q_0 \in \mathbb{Z}_{(2)}$, and $q - q_0 \in 2^Q\mathcal{O}$, then for every integer $0 \le m < 2^Q$,*

$$\nu_2\left(\prod_{k=0}^{m-1}(q-k)\right) \ge \nu_2(m!).$$

Proof. For $1 \le r \le \lfloor \log_2 m \rfloor$, the rational q_0 has a well-defined integer residue modulo 2^r. Among m consecutive integers, at least $\lfloor m/2^r \rfloor$ have that residue. Since $r \le Q$, the corresponding factors $q-k$ are divisible by 2^r. Summing over r proves the result by Legendre's formula. For $m = 0, 1$ the assertion is immediate. □

This lemma is necessary: an arbitrary extension element of $\mathcal{O}$ need not have integral binomial coefficients. The actual rows have the required proximity to the prime subring.

Lemma 7.3 (Valuation of the pivot product)**.** *For every actual crossing row and every $0 \le b < J$,*

$$\nu_2\left(\prod_{h=0}^{b-1}(w_x - v_h)\right) \ge \eta_b.$$

Furthermore, η_b is nondecreasing and

$$\eta_b = \frac{22}{21}b + O(\log(b+2)).$$

Proof. Suppose first that $R \ge 3$. The permitted residues split into three cases. If w_x is odd, the A factors give

$$2^{A_b}\prod_{k<A_b}\left(\frac{w_x-1}{2} - k\right),$$

while every B or C factor is a unit. Their valuation is at least $A_b + \nu_2(A_b!)$.

If $w_x \equiv 0 \pmod 4$, the B factors similarly contribute at least $2B_b + \nu_2(B_b!)$, each C factor contributes exactly one power of two, and the A factors are units. The resulting bound is the second term in (22).

The remaining permitted case is $w_x \equiv c \pmod 8$. Now the C factors contribute at least $3C_b + \nu_2(C_b!)$, each B factor contributes one power of two, and the A factors are units. This gives the third term.

To justify the factorial steps over $\mathcal{O}$, apply Lemma 7.2 after dividing by two, four, or eight. Lemma 7.1 supplies precisions $R-1$, $R-2$, and $R-3$ respectively. The first two factor counts satisfy

$$A_b \le b < J = 2^{R-2} < 2^{R-1}, \qquad B_b \le b < 2^{R-2}.$$

If $b = 21q + r$, $0 \le r \le 20$, the positions of the letter C give

$$C_b = 4q + \lfloor r/5 \rfloor \le \lfloor b/5 \rfloor \le \lfloor b/4 \rfloor.$$

Thus, whenever $C_b > 0$, $C_b < 2^{R-3}$. If $R \le 3$, no C factor occurs. The case $R = 2$ has $b = 0$, so its product bound is immediate.

Each expression in the minimum (22) is nondecreasing as the three counts increase. For $b = 21q$, Legendre's formula gives the exact identity

$$\eta_{21q} = 22q - \max\{s_2(11q), s_2(6q), s_2(4q)\},$$

where s_2 is the number of ones in the binary expansion. For general b, the three counts differ by $O(1)$ from $(11b/21, 6b/21, 4b/21)$. The bound $s_2(m) = O(\log(m+2))$ proves the claimed asymptotic formula. □

7.3 A finite separation with no nonunit division

For arbitrary even auxiliary points ρ_h, set

$$U_b(z) = \prod_{h<b} \frac{z-\rho_h}{1-hz}, \qquad V_b(j) = \prod_{h<b} \frac{j-h}{1-\rho_h j}.$$

All denominators are units for $z \in 2\mathcal{O}$ and integer j.

Lemma 7.4 (Finite Cauchy elimination)**.** *For every nonnegative integer q,*

$$\frac{1}{1-zj} = \sum_{b<q} \frac{(1-b\rho_b)U_b(z)V_b(j)}{(1-bz)(1-\rho_b j)} + \frac{U_q(z)V_q(j)}{1-zj}. \tag{23}$$

If $0 \le j < J$ and $q = J$, the remainder is zero.

Proof. The elementary identity

$$(1-bz)(1-\rho_b j) - (z-\rho_b)(j-b) = (1-b\rho_b)(1-zj)$$

shows that the bth summand is the difference of the consecutive remainders $U_b(z)V_b(j)/(1-zj)$ and $U_{b+1}(z)V_{b+1}(j)/(1-zj)$. Telescope. For $j < J$, the product $V_J(j)$ contains the zero factor $j-j$. □

For the specific points (21), define

$$\ell_b(j) = \frac{(1-b\rho_b)\binom{j}{b}}{\prod_{h=0}^{b}(1-\rho_h j)}, \tag{24}$$

$$r_b(x) = c_x\, b! \,\frac{\prod_{h=0}^{b-1}(z_x-\rho_h)}{\prod_{h=0}^{b}(1-hz_x)}. \tag{25}$$

Here $\binom{j}{b} = 0$ for $b > j$.

Theorem 7.5 (Residue-aware separated kernel)**.** *For every actual crossing entry,*

$$W_{xy}(T) = \sum_{b=0}^{J-1} r_b(x)\ell_b(j).$$

Every source feature $\ell_b(j)$ belongs to $\mathbb{Z}_{(2)}$. The target coefficient satisfies

$$\nu_2(r_b(x)) \ge \omega_b, \qquad \omega_b = 1 + b + \nu_2(b!) + \eta_b = \frac{64}{21}b + O(\log(b+2)). \tag{26}$$

The weights ω_b are strictly increasing. The stronger offset R may replace one in this formula.

Proof. Use Lemma 7.4 with $q = J$, and replace $\prod_{h<b}(j-h)$ by the integer product $b!\binom{j}{b}$. The integer binomial and the odd denominators make (24) integral. In (25), factoring two from each of the b numerator factors gives $2^b \prod_{h<b}(w_x - v_h)$. Combine Lemma 7.3 with the explicit $b!$ and $\nu_2(c_x) \ge 1$. Legendre's formula gives the slope $1 + 1 + 22/21 = 64/21$. Finally, η_b is nondecreasing and the term b increases strictly, so the weights do also. □

The formula multiplies the factorial into the target. It never divides by a factorial in the coefficient ring. Implementation uses integer binomial coefficients, ring multiplication, and unit inversion only. The points ρ_h are auxiliary interpolation points; they need not be actual rows in a packet.

7.4 Consequences for the contraction accounting

For each existing column tile, replace its source moment of degree b by

$$L_b(X) = \sum_{\substack{\text{real columns } y \text{ in the tile} \\ y=y_0+2^{i+1}j}} \ell_b(j) X_y.$$

There is still one mode of each weight per old tile. It is a linear form in bits of the same source group and targets the same other group. After expanding the row factors, a mode-exponent tuple β whose total certified weight is at least the working precision p has coefficient zero modulo 2^p. Thus the retained condition is

$$\sum_{\text{tile families } f} \sum_{b=0}^{J-1} \omega_b \beta_{f,b} < p.$$

The source's child factorization, wire count, degree bounds, and creation-precision convention are preserved. Products defining the features contain at most $J \le L$ factors of polynomial description length, so the entry-oracle bounds remain valid. This argument also permits the strongest compatible previous packet-entry routine.

To quantify the gain, let $N_\gamma(c,p)$ denote the abstract tuple count with c families and weights $\omega_b = \gamma b + O(\log(b+2))$. Its standard generating-product bound is

$$N_\gamma(c,p) \le \exp\left(\theta p + c \sum_{b \ge 0} -\ln(1 - e^{-\theta\omega_b}) \right).$$

As $\theta \downarrow 0$, the sum has leading term $\pi^2/(6\gamma\theta)$. Optimizing θ gives the leading logarithmic bound

$$\ln N_\gamma(c,p) \le \left(\pi \sqrt{\frac{2}{3\gamma}} + o(1) \right) \sqrt{cp}.$$

The actual finite cutoff only reduces this bound. In the active-call regime, $p/c \to \infty$, so the displayed asymptotic applies uniformly with the previously verified schedule.

Consequently, replacing the previous factorial-Cauchy weights of slope 3 by (26), while retaining a matched precision and recursion schedule, multiplies the leading enumeration coefficient by

$$\sqrt{\frac{3}{64/21}} = \sqrt{\frac{63}{64}} = 0.9921567416\ldots. \tag{27}$$

Relative to the geometric weights of slope one, the factor is $\sqrt{21/64}$. These are comparisons of explicit upper-bound certificates. They are not wall-clock speedup measurements or lower bounds on the old algorithm, and they do not change the logarithmic order of the global SAT bound.

The new weights need not dominate the old weights at every finite index. For example, the new weight at $b = 4$ is 10, while the old factorial-Cauchy weight is 11. The improvement is in the proved leading asymptotic coefficient. If desired, an implementation may count the old and new abstract tuple sets by ordinary generating-function dynamic programming and choose the smaller set at each call. The numbers of families, modes and precision units are polynomial in s, and the counts have polynomial bit length, so this selection has polynomial bit cost and space. It preserves both exact identities and the new asymptotic upper bound. No uniform finite-input speedup is asserted.

8 Entropy and the complete contraction theorem

Suppose the common mode weights have the form

$$w_b = \gamma b + O(\log(b+2)), \qquad w_b \geq 1,$$

and are strictly increasing. The new kernel in the preceding section gives $\gamma = 64/21$. The old factorial-Cauchy basis gives $\gamma = 3$. There are at most $c_t = (2+o(1))M/D_i$ families. Extending each finite family to infinitely many formal weights only enlarges the count.

For $\theta \downarrow 0$, comparison with the integral $\int_0^\infty -\ln(1-e^{-x})\,dx = \pi^2/6$ gives

$$\sum_{b\geq 0} -\ln(1-e^{-\theta w_b}) = \frac{\pi^2}{6\gamma\theta} + o(\theta^{-1}). \tag{28}$$

One way to justify the perturbation is to fix any $\eta > 0$: for all sufficiently large b, $(\gamma-\eta)b \leq w_b \leq (\gamma+\eta)b$. The finitely many omitted indices contribute $O(\ln(1/\theta))$. The integral comparison squeezes θ times the sum between $\pi^2/(6(\gamma+\eta))$ and $\pi^2/(6(\gamma-\eta))$; let $\eta \downarrow 0$. This proves the precise leading term without requiring an unproved partition formula.

If $\mathcal{N}_t$ counts tuples satisfying $\sum_{f,b} w_b\beta_{f,b} < P_t$, the generating product yields

$$\ln \mathcal{N}_t \leq \theta P_t + c_t \sum_{b\geq 0} -\ln(1-e^{-\theta w_b}).$$

Use $P_t \leq (6+o(1))MD_i/u_d$ and $c_t \leq (2+o(1))M/D_i$, and optimize the two leading terms. The saddle scale is a constant times $\sqrt{u_d}/D_i$, which tends uniformly to zero by (19). We obtain

$$\ln \mathcal{N}_t \leq \left(2\pi\sqrt{\frac{2}{\gamma}} + o(1)\right)\frac{M}{\sqrt{u_d}}. \tag{29}$$

Integer ceilings on family counts are negligible because M/D_i tends uniformly to infinity. The saddle estimate and the preceding perturbation argument are therefore uniform across the recursion.

Theorem 8.1 (Explicit refined contraction)**.** *Assume the normalized interface of Section 3 and the unchanged integral deformation and child identities of the source. Use the exact taper (5), precision (7), schedule (14), sharp packing, and the proved integral mode basis of slope $\gamma = 64/21$. Then deterministic exact contraction uses polynomial space and at most*

$$(s+\mathcal{L})^{O(1)}2^{(C_\gamma+o(1))s/\sqrt{\lambda}} \quad \textit{bit operations}, \qquad C_\gamma = \frac{9\pi}{4\ln 2}\sqrt{\frac{520}{\gamma}}.$$

In particular $C_{64/21} = 133.20737249245\ldots$. The same new packing and schedule with the existing slope-3 basis gives $C_3 = 134.2604115868\ldots$.

Proof. Correctness follows by the same induction from leaf contractions as in the source. The new mode identity is exact, all omitted coefficients vanish at the working precision, and the source/target factorization is unchanged. Hermite recovery has enough jets by (4), at sufficient precision by (7). Creation-time lifts remain fixed.

For cost, (17) and $u_d = (1+o(1))\lambda/(260\cdot 9^d)$ give

$$\sum_{d \text{ active}} \frac{M_d}{\sqrt{u_d}} \leq (1+o(1))\sqrt{260}\frac{s}{\sqrt{\lambda}}\sum_{d\geq 0}(3/27)^d = (1+o(1))\frac{9}{8}\sqrt{260}\frac{s}{\sqrt{\lambda}}.$$

Combining this with (29), and dividing the natural logarithm by $\ln 2$, gives the asserted coefficient. Node choices, interpolation values and group choices give only polynomial branching per call, so their accumulated logarithm is $O(\lambda \log \lambda) = o(s/\sqrt{\lambda})$. A depth-first leaf sums over at most $H = O(s/\lambda)$ bits. Its direct enumeration and entry evaluation have logarithmic costs $O(s/\lambda) + O(s^{9/10}) = o(s/\sqrt{\lambda})$.

Arithmetic, packing and local descriptions have polynomial cost per generated term outside the permitted entry work. Because the branching bound is independent of $\mathcal{L}$, dependence on description length stays a fixed polynomial rather than a depth-dependent polynomial power. The finite Cauchy features require only polynomially many exact ring operations and preserve the same packet-entry bound.

For space, enumerate one tuple and one interpolation value at a time, accumulate a product of child values and then the running sum, and keep only the current depth-first stack. There are $O(\log \lambda)$ levels. Each holds polynomially many polynomial-bit data, immutable descriptions, and residues. Exact rational recovery is the source's polynomial-bit calculation. Thus space remains polynomial in $s + \mathcal{L}$. □

8.1 Why these schedule constants were chosen

More generally, retain the old variance bound, use maximum degree s^{δ} for fixed $\delta < 1$, and set $u_d = \left\lfloor \varepsilon\lambda/b^d \right\rfloor$ with the same additive guard. Provided $1 < b < r/2$ and

$$\varepsilon < \frac{(1-\delta)(b-1)}{4b \log_2 r},$$

the leading coefficient is

$$\frac{2\pi}{\ln 2} \frac{\sqrt{2/(\gamma\varepsilon)}}{1 - (2/r)\sqrt{b}}. \tag{30}$$

At the boundary of the available packet budget, differentiation in b gives $b = (r/2)^{2/3}$. The resulting infimum within this certificate family is

$$\frac{2\pi}{\ln 2} \frac{\sqrt{8 \log_2 r/(\gamma(1-\delta))}}{(1 - (2/r)^{2/3})^{3/2}}.$$

The continuous stationary equation in r is $(r/2)^{2/3} = 1 + 2\ln r$, near $r = 54$. Our clean integers $r = 54$, $b = 9$, $\varepsilon = 1/260$ lie strictly inside the allowed region for $\delta = 9/10$. One also requires $\varepsilon b/(b-1) < \delta$ so that new packet vertices respect the degree cap; this holds for every numerical tuple used here. This explains the choice without claiming global optimality among contraction algorithms.

9 Consequences for 3-SAT and density-sensitive inputs

For fixed clause density the direct clause/equality network has $s = O_k(n+m) = O_{k,D}(n)$. For arbitrary density the source's parity-query and incidence-index reduction has $s = O_k(n \log\log n)$. Substitute these sizes in Theorem 8.1; all surrounding overheads are lower order in the corresponding exponential scale or part of the polynomial factor. This improves the specified contraction certificate in both reductions and preserves polynomial space.

A numerical coefficient of $133.207\ldots$ must not be written directly in front of n or $n \log\log n$ without tracking the constant in the reduction's wire budget. The theorem is expressed in the normalized network size precisely to avoid that ambiguity.

The existing scalar parity refinement can be kept at the reduction stage. It removes a truncated polynomial coordinate and reduces a retained incidence-degree bound, while our theorem permits $\ell = 1$.

The two density packets can likewise keep their row-selection procedures and their $O(n \log(2 + B))$ wire budget for $B \le \log^A n$ with fixed A. The new contraction then gives the same density-sensitive order with the improved normalized certificate. Neither composition creates a new arbitrary-density log-power by itself.

9.1 A further formula-specific degree choice

The sparse-query proof groups tests and original variables separately into $J = \left\lceil n^{1/4} \right\rceil$ groups. This exponent is not essential. For example choose $J = \lceil n^{0.495} \rceil$. The total-wire estimate remains $O(n \log(2 + S))$ when each query has at most S monomials and $S = \log^{O(1)} n$: the extra $J^2 = n^{0.99+o(1)}$ rounding term is $o(n)$. Degree-array enumeration costs $2^{O(n^{0.99} \log n)}$, which is lower order than $2^{\Theta(n/\sqrt{\log n})}$. After the source's high-incidence branching, each group has degree $n^{0.505} \log^{O(1)} n \le s^{0.51}$ eventually.

The original entry-evaluation proof continues to apply. A test-group entry is a product of independent per-test finite-difference coefficients, each over a polylogarithmic scope. A variable-group entry is the power of two determined by the number of untouched variables. Increasing the number of groups does not create an exponential product of entry runtimes [3].

Consequently the same proof can use $\delta = 0.51$ and $\varepsilon = 1/54$, with $r = 54$ and $b = 9$. Here $\sum u_d \le \lambda/48$, and

$$0.51 + \frac{\log_2 54}{12} < 1$$

supplies strict packing slack. Formula (30) then gives the smaller network coefficient

$$C_{\text{query}} = \frac{9\pi}{4 \ln 2} \sqrt{\frac{108}{64/21}} = 60.7069575490\ldots. \tag{31}$$

The exact decimal is generated by the included scripts; the displayed formula is authoritative. This secondary result requires the lower degree promise and the stated regrouping. It is not the apples-to-apples comparison against the generic $s^{9/10}$ interface in Table 1. Both have the same log-power.

10 An additional exact variance lemma

The sharp-padding result already proves the main theorem using the source's variance bound. A further structural refinement is available when parallel-edge multiplicity is controlled.

Lemma 10.1 (Exact collision variance)**.** *Give vertices on each side independent uniform labels from a set of size R. Let P be the number of ordered edge pairs with both endpoints in common, including each edge paired with itself, and let S count ordered edge pairs sharing exactly one endpoint. If X_γ is the load of a pair of labels, then*

$$\mathbb{E} \sum_{\gamma} (X_\gamma - M/R^2)^2 = P(1 - R^{-2}) + S(R^{-1} - R^{-2}).$$

In particular, if maximum degree is Δ and maximum parallel multiplicity is μ, the expectation is at most $M\mu + 2M\Delta/R$.

Proof. Expand the sum as the number of ordered edge pairs receiving identical label pairs, minus M^2/R^2. For disjoint endpoints the agreement probability is R^{-2}, for exactly one common endpoint it is R^{-1}, and for two it is one. Subtracting R^{-2} per pair gives the equality. Also $P = \sum_{u,v} \mu_{uv}^2 \le M\mu$ and $S \le 2M\Delta$, which give the upper bound. □

With $R = r^a$, the corresponding relative padding error is bounded by

$$\sqrt{\frac{\mu R^4 + 2\Delta R^3}{M}} + \frac{LR^2}{M}.$$

Thus $\mu R^4 = o(M)$ and $\Delta R^3 = o(M)$ can replace the older $\Delta R^4 = o(M)$ condition. A child has parallel multiplicity at most $2L$ because one endpoint of every newly formed edge is a packet vertex. A generic root, however, can have much larger parallel multiplicity. We do not silently assume a simple root or use this stronger variance estimate in the primary coefficient. It is a separately proved improvement available to future instances with an authenticated multiplicity bound.

11 What remains unresolved

The new weight sequence is still linear: $w_b = (64/21)b + O(\log b)$. Therefore its partition generating function still has logarithm proportional to $1/\theta$. Combining it with precision $P_t = \Theta(MD_i/u_d)$ still gives a square-root saddle, namely $M/\sqrt{u_d}$. Packing and the guard improve the constants and how the available logarithmic budget is spent; they do not turn that saddle into M/u_d.

The old archive's binary-selection argument also remains relevant: an abstract inventory with $\Theta(M/D)$ linear-weight modes has $2^{\Omega(M/\sqrt{u})}$ feasible tuples within an appropriate budget. This is a barrier to improving the entire tuple-universe count using only that numerical envelope, not a lower bound for SAT or for non-enumerative contraction [10, 4]. The new residue restriction changes the linear coefficient of the weights and therefore escapes no such order-level barrier.

The dense-input $\log \log n$ has a separate origin. It encodes degree-$O(n)$ subsets of an incidence universe of size n polylog n. Source and target group tensors remember which test supplied an incidence, so replacing that code with the union of original variables requires a new aggregation argument. The previous degree-preserving compression, structured-mode summation, and second-contraction proposals isolate this missing structure, but none of the local improvements in this report constructs it.

For the distinct 49-page crossing-decomposition algorithm, a stronger log-power would follow from admissible pieces with a smaller contraction factor ρ, together with all uniform description and evaluation bounds required in its Definition 7. Its current partition-based construction has $\rho = 1/\sqrt{2}$. The present two-adic basis is not automatically a set of such pieces. No unproved transfer between the two architectures is used.

Scope of novelty. The elementary finite Cauchy telescope is classical in character and already appears in the factorial-Cauchy packet. The new use here is the actual-row residue exclusion, the explicit 11:6:4 pivot allocation, and its compatible valuation gain. Sharp padding and the additive guard were not present in the seven reviewed packets. We establish novelty relative to that inspected corpus, not a literature-wide priority claim.

12 Executable artifacts and verification

The bundle contains the complete LaTeX source, its compiled PDF, standard-library Python programs, recorded outputs, and a source/commit manifest. The programs check finite exact statements underlying the new lemmas. They are not a complete implementation of the preprint's SAT counter. No SAT benchmark, complete Lean build, or transitive axiom audit is claimed.

Exact finite check	Cases
Pivot-prefix counts, monotonicity and weight estimates	10,000
Integer-domain pivot-product valuation inequalities	8,505
Actual crossing-row parameter and residue cases	19,008
Included cases with non-prime-field interpolation	9,504
Exact finite Cauchy separation identities	50,496
Target-coefficient valuation inequalities	239,808
Schedule guard and packet-budget identities	25,704
Selected logarithmic sizes, with all provisional active depths	9
Small multigraph/label-size configurations	160
Fully enumerated vertex labelings	6,096
Finite abstract mode inventories	21

The kernel, schedule and packing checks pass with ordinary Python and with assertions disabled by `python3 -O`; their guards use explicit exceptions. The result records preserve the scope and the code hashes. The finite mode inventories use one, two or four colors and strict budgets from 16 to 1024. In all 21 of these sampled inventories the old factorial-Cauchy sequence has fewer admissible tuples than the new periodic sequence. This is consistent with the finite-index qualification in Section 7: the theorem is asymptotic, and the periodic sequence has lower-order costs. The optional exact-count selection keeps the old basis whenever its finite inventory is smaller. These data are included as a check against interpreting the asymptotic coefficient as a uniform finite-input speedup.

The mode checks use exact rational arithmetic in $\mathbb{Q}[U]/(U^2+U+1)$, including $\zeta = U$ and $1+U$. These non-prime-field interpolation values matter: ordinary binomial integrality is false for arbitrary elements of an unramified extension. The proof and checks instead use the actual rows' high-precision congruence to the prime subring.

To rerun the checks, execute `python3 code/run_checks.py` from the extracted archive. To rebuild the PDF, run `pdflatex -no-shell-escape -halt-on-error report.tex` twice from the archive root. Standard AMS, Latin Modern, geometry, microtype, mathrsfs, needspace, hyperref, xurl, and table packages are needed. The check suite itself requires only Python's standard library and uses no network. The source manifest identifies the reviewed external files without executing or bundling their programs.

13 Research outcome

The deliverable establishes an explicit improvement in the exponential certificate of the reviewed OpenAI-style contraction, with fully stated new local lemmas and preserved recursive interfaces. The clean generic coefficient is 133.20737249245 . . .; under the explained lower-degree sparse-query interface a further formula-specific coefficient follows. The generic result already improves the strongest comparable previous certificate, including its straightforward composition with factorial Cauchy modes.

The stronger goal of improving $2^{O(n \log\log n/\sqrt{\log n})}$ to a strictly smaller exponent order remains open in this work. The exact point of progress is therefore a proved quantitative refinement, not a completed new log-power theorem.

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